Gap-Aware Retrieval (GAR):

Adaptive Retrieval Using Information Gaps and Existing Evidence

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ABSTRACT

Retrieval-Augmented Generation (RAG) systems use external documents to provide large language models (LLMs) with information that may not be available in their parameters. In iterative and multi-hop information-seeking tasks, however, a conventional retriever may repeatedly return documents that are relevant to the original question but do not resolve the information that remains missing.

This project develops a methodology for investigating **Gap-Aware Retrieval (GAR)** within the Missing Information Guided Retrieve-Extraction-Solving (MIGRES) framework. MIGRES supplies the current question, previously collected evidence, and the information gap at each retrieval iteration. GAR uses these signals to rank candidate documents according to gap-aware relevance, coverage of missing information, and novelty with respect to existing evidence.

The methodology uses a fixed sentence-embedding model, approximate nearest-neighbor (ANN) candidate generation, and a weighted GAR scoring function. The proposed retriever is evaluated against BM25, DPR, Contriever, ColBERTv2, and SPLADE under controlled experimental conditions, with the validation split used for selecting GAR scoring hyperparameters before the test evaluation.

The evaluation considers retrieval effectiveness, downstream question-answering quality, ablation of the GAR components, computational efficiency, and reproducibility. The methodology is designed to determine whether explicitly modeling the information gap and previously collected evidence improves iterative retrieval without changing the surrounding MIGRES reasoning, extraction, and solving components.

Keywords: Retrieval-Augmented Generation, Information Retrieval, MIGRES, Gap-Aware Retrieval, Information Gap, Multi-Hop Question Answering, Dense Retrieval, Approximate Nearest Neighbor Search.

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CHAPTER 1

INTRODUCTION

Retrieval-Augmented Generation (RAG) combines information retrieval with large language models (LLMs) to improve answer generation using information from external documents. Instead of relying only on the knowledge stored in the model, RAG retrieves relevant information and provides it as context to the LLM. The quality of the retrieved information therefore has an important effect on the final answer.

Traditional retrieval methods mainly select documents based on their relevance to the given query. This can become challenging for complex and multi-hop questions, where the answer may require information from multiple documents. After one retrieval step, some information may already be available while other information is still missing. In such cases, retrieving documents based only on the original query may result in repeated or less useful information.

The MIGRES framework addresses this issue by using missing information to guide iterative retrieval. After recovering and processing the evidence, MIGRES identifies what information is still missing and uses it to guide the next retrieval step [1]. This provides a useful direction for improving retrieval in multi-step question answering.

Recent research has also explored other ways to improve the usefulness of the retrieved information. Relevant Information Gain considers whether retrieved information provides useful new content rather than unnecessary repetition [2]. InfoGain-RAG similarly investigates document information gain for reranking and filtering retrieved documents [3]. Other work has considered source reliability as an additional factor in RAG [4], while ColBERTv2 provides an efficient late-interaction approach for document retrieval [5].

From the study of MIGRES and these related approaches, this project identified a possible opportunity to improve iterative retrieval. The main observation is that the retrieval process could consider not only the original question but also what information is currently missing and what evidence has already been collected.

The study of MIGRES and related retrieval approaches has motivated further investigation into how iterative retrieval can be improved. This project will explore the idea of **Gap-Aware Retrieval (GAR)**, focusing on whether the current information gap and previously collected evidence can help guide iterative retrieval. The concept will be further studied, refined, and developed during the subsequent stages of the project.

CHAPTER 2

LITERATURE SURVEY

Table 2.1: Literature Survey 1

Name of paper	**LLMs Know What They Need: Leveraging a Missing Information Guided Framework to Empower Retrieval-Augmented Generation**
Authors	Keheng Wang, Feiyu Duan, Peiguang Li, Sirui Wang, and Xunliang Cai
Publication Details	Proceedings of the 31st International Conference on Computational Linguistics (COLING), 2025, pp. 2379–2400.
Research Findings	• The ability of LLMs to extract useful information from retrieved documents and identify information that is still missing is experimentally demonstrated. • MIGRES is introduced as a Missing Information Guided Retrieve-Extraction-Solving framework, where the identified missing information is used to generate a targeted query for subsequent retrieval. • Sentence-level reranking and filtering are introduced to remove irrelevant content from retrieved documents before useful information is extracted.
Research Gap	• Although MIGRES improves iterative retrieval, its performance still depends on the quality of the generated queries and the knowledge returned by the retriever. • The framework can still encounter hard-negative passages that receive high relevance scores but do not provide useful information, which makes effective evidence selection challenging.
Relevance with Our Work	• MIGRES provides the base iterative RAG framework for investigating whether the retrieval stage can be further improved using the information available at each retrieval step.

Table 2.2: Literature Survey 2

Name of paper	**Better RAG using Relevant Information Gain**
Authors	Marc Pickett, Jeremy Hartman, Ayan Kumar Bhowmick, Raquib-ul Alam, and Aditya Vempaty
Publication Details	arXiv preprint, 2024.
Research Findings	• The limited context window of RAG systems can cause redundant passages to occupy space that could otherwise be used for useful information. • A Relevant Information Gain metric is introduced to measure the total information relevant to a query provided by a set of retrieved passages. • The Dartboard retrieval method is developed using the proposed metric, allowing diverse passages to be selected without explicitly adding a separate diversity term. • The method is evaluated on the Retrieval Augmented Generation Benchmark (RGB) and is reported to outperform existing retrieval approaches based on relevance and diversity.
Research Gap	• The proposed retrieval objective is defined for selecting a set of passages for a query, while the study does not specifically address how the retrieval objective should change when new evidence is collected through multiple retrieval steps. • The evaluation is mainly conducted using the RGB benchmark, leaving the effectiveness of the approach in other iterative or multi-step retrieval settings open for further investigation.
Relevance with Our Work	• The paper provides a basis for considering the usefulness and redundancy of retrieved information when selecting documents, which is relevant to the retrieval problem being investigated in this project.

Table 2.3: Literature Survey 3

Name of paper	**InfoGain-RAG: Boosting Retrieval-Augmented Generation through Document Information Gain-based Reranking and Filtering**
Authors	Zihan Wang, Zihan Liang, Zhou Shao, Yufei Ma, Huangyu Dai, Ben Chen, Lingtao Mao, Chenyi Lei, Yuqing Ding, and Han Li
Publication Details	Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, pp. 7190–7204.
Research Findings	• Document Information Gain (DIG) is introduced to measure the contribution of an individual retrieved document to correct answer generation. • DIG is calculated from the change in the LLM's generation confidence when a document is added to the context. • InfoGain-RAG uses DIG scores to train a multi-task reranker that performs document filtering and ranking. • Experiments across different RAG models and benchmarks show that the proposed method improves answer generation compared with several existing retrieval and reranking approaches.
Research Gap	• The current implementation is evaluated only on text-based retrieval tasks, while its application to other modalities such as visual or code information is left for future work. • The DIG metric measures the contribution of a document to answer generation but cannot determine whether the information within that document is factually correct. • The reranker size is limited by computational constraints, which may restrict the performance that could be obtained from larger models.
Relevance with Our Work	• The paper provides a useful basis for considering the contribution of retrieved documents when designing an improved document selection strategy.

Table 2.4: Literature Survey 4

Name of paper	**Retrieval-Augmented Generation with Estimation of Source Reliability**
Authors	Jeongyeon Hwang, Junyoung Park, Hyejin Park, Dongwoo Kim, Sangdon Park, and Jungseul Ok
Publication Details	Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2025, pp. 34279–34303.
Research Findings	• The effect of heterogeneous source reliability in multi-source RAG is investigated. It is shown that relying only on query-document relevance can allow information from unreliable sources to influence the generated answer. • Reliability-Aware RAG (RA-RAG) is introduced, where source reliability is estimated by cross-checking information across multiple sources. • Documents are selectively retrieved from highly reliable and relevant sources, and their information is aggregated using Weighted Majority Voting (WMV). • Experiments show that RA-RAG improves performance over the compared baselines and remains effective as the number of information sources increases.
Research Gap	• Reliability estimation becomes more difficult when only a limited number of sources are available for cross-verification. • The evaluation of real-world source reliability is mainly focused on news-related claims, while specialized topics with fewer available reference sources remain challenging. • The reliability estimation process is performed in an offline setting and therefore requires periodic updates when the reliability of a source changes over time.
Relevance with Our Work	• The paper demonstrates that factors beyond query-document relevance can be incorporated into document selection, supporting the investigation of additional evidence-aware retrieval factors.

Table 2.5: Literature Survey 5

Name of paper	**ColBERTv2: Effective and Efficient Retrieval via Lightweight Late Interaction**
Authors	Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia
Publication Details	Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL), 2022, pp. 3715–3734.
Research Findings	• ColBERTv2 uses a late-interaction architecture in which query and document representations are independently encoded and compared at the token level. • A residual compression mechanism is introduced to reduce the storage requirements of token-level document representations. • Denoised supervision is used to improve the training of the retriever by reducing the effect of false negatives in the training data. • ColBERTv2 achieves strong retrieval effectiveness while requiring substantially less storage than the original ColBERT model.
Research Gap	• The method still requires storing token-level representations for documents, which can result in considerable storage requirements for large collections. • The retrieval architecture is designed primarily for efficient query-document matching, while the study does not investigate iterative retrieval with changing information requirements.
Relevance with Our Work	• ColBERTv2 provides an established neural retrieval method that can be considered as a retrieval baseline for evaluating an alternative retrieval strategy.

CHAPTER 3

PROBLEM DEFINITION

1. In iterative RAG, conventional retrieval may repeatedly retrieve documents that are relevant to the original query but contain information that has already been obtained. This can lead to redundant retrieval and reduce the usefulness of later retrieval steps.
2. The information required to answer a multi-hop question can change as new evidence is collected. Existing retrieval methods may not sufficiently use the current missing information when selecting documents for the next retrieval step.
3. Previously retrieved evidence can contain information that is already available to the system. A retrieval mechanism is therefore needed to consider the collected evidence along with the remaining information requirement when selecting further documents.
4. The effectiveness of using the current information gap and previously collected evidence for iterative retrieval needs to be experimentally investigated and compared with existing retrieval approaches.

CHAPTER 4

OBJECTIVES

1. To investigate methods for reducing redundant retrieval of information that has already been obtained.
2. To study and incorporate the current missing information as a factor in selecting documents during iterative retrieval.
3. To investigate how previously collected evidence can be considered when selecting documents for subsequent retrieval steps.
4. To evaluate the effectiveness of the developed retrieval approach against existing retrieval methods using suitable retrieval and question-answering measures.

CHAPTER 5

SOLUTION STRATEGY / METHODOLOGY

The solution strategy improves iterative document retrieval by considering not only the relevance of a document to the question, but also the information that is still missing and the evidence already collected. The approach consists of four main strategies:

1. **Reduce redundant retrieval:** Prioritize documents that provide new information rather than repeating previously retrieved content.
2. **Focus on missing information:** Use the current information gap to prioritize documents that address the information still required.
3. **Consider existing evidence:** Use previously collected evidence when ranking documents for subsequent retrieval.
4. **Evaluate retrieval performance:** Compare the proposed approach with established retrieval methods using consistent datasets and evaluation settings.

5.1 Proposed Solution

The proposed solution is Gap-Aware Retrieval (GAR), an iterative retrieval approach designed to improve how documents are selected when information is collected over multiple retrieval steps. At each iteration, GAR considers the original question, the information that is still missing, and the evidence that has already been collected. This allows the retrieval process to move beyond simply finding documents that are relevant to the question and instead focus on documents that can provide useful and new information.

GAR first represents the question, missing information, documents, and collected evidence using a common semantic representation. It then uses Approximate Nearest Neighbor(**ANN**) search to identify a manageable set of candidate documents. These candidates are ranked using three signals: their relevance to the question and current information gap, the amount of missing information they can cover, and the amount of new information they provide compared with existing evidence. The highest-ranked documents are selected for the next retrieval step, after which the updated information state can be used for further retrieval. This process allows GAR to progressively improve the retrieved evidence while reducing unnecessary repetition.

5.1.1 Architecture Diagram

The architecture shows how GAR operates within the MIGRES framework. MIGRES provides the current information gap and accumulated evidence, while GAR generates candidate documents and ranks them using relevance, gap coverage, and evidence novelty.

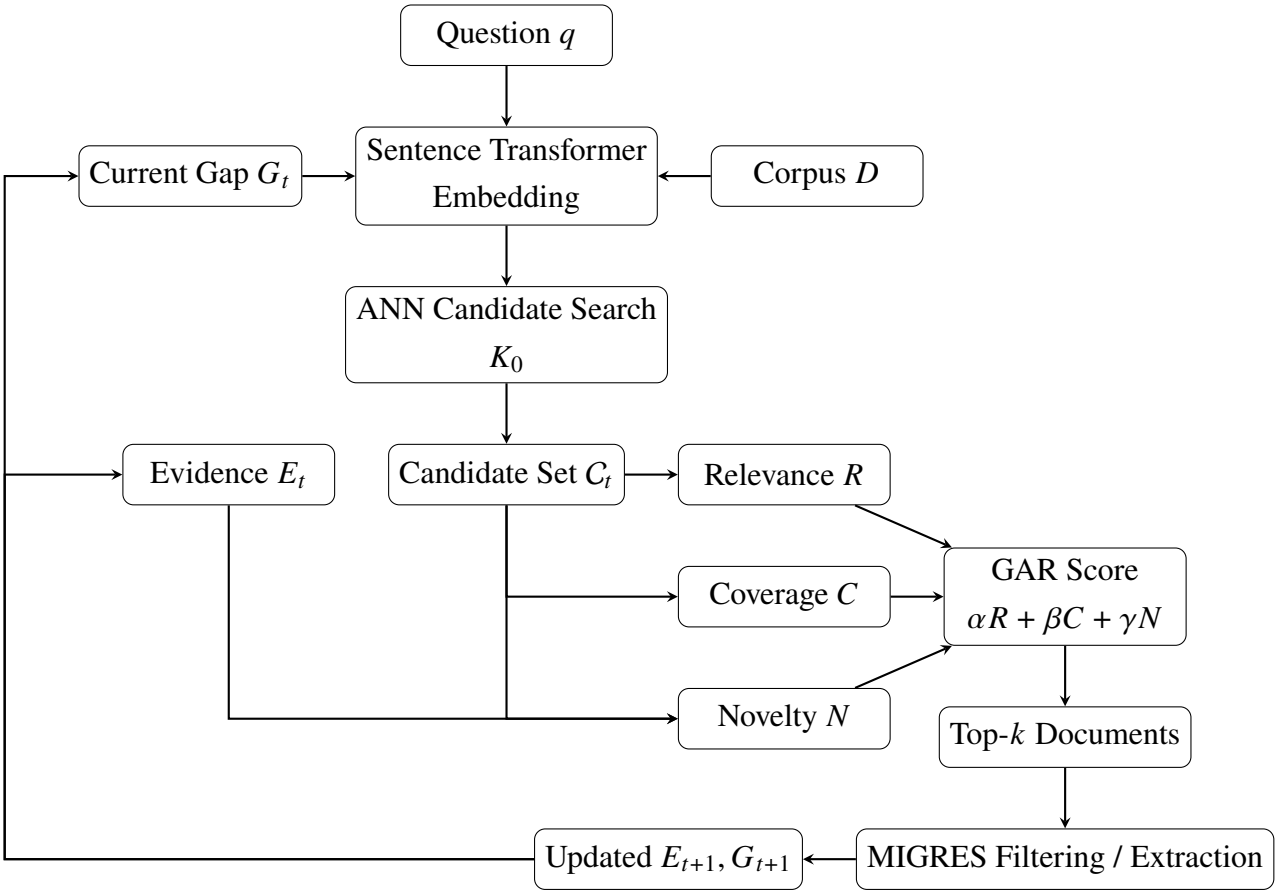


Figure 5.1: GAR methodology within the MIGRES framework.

5.1.2 Flow Diagram

The flow diagram presents the iterative execution of the proposed method. After each retrieval and evidence-extraction step, MIGRES updates the information state. GAR then performs the next retrieval until the stopping condition is satisfied.

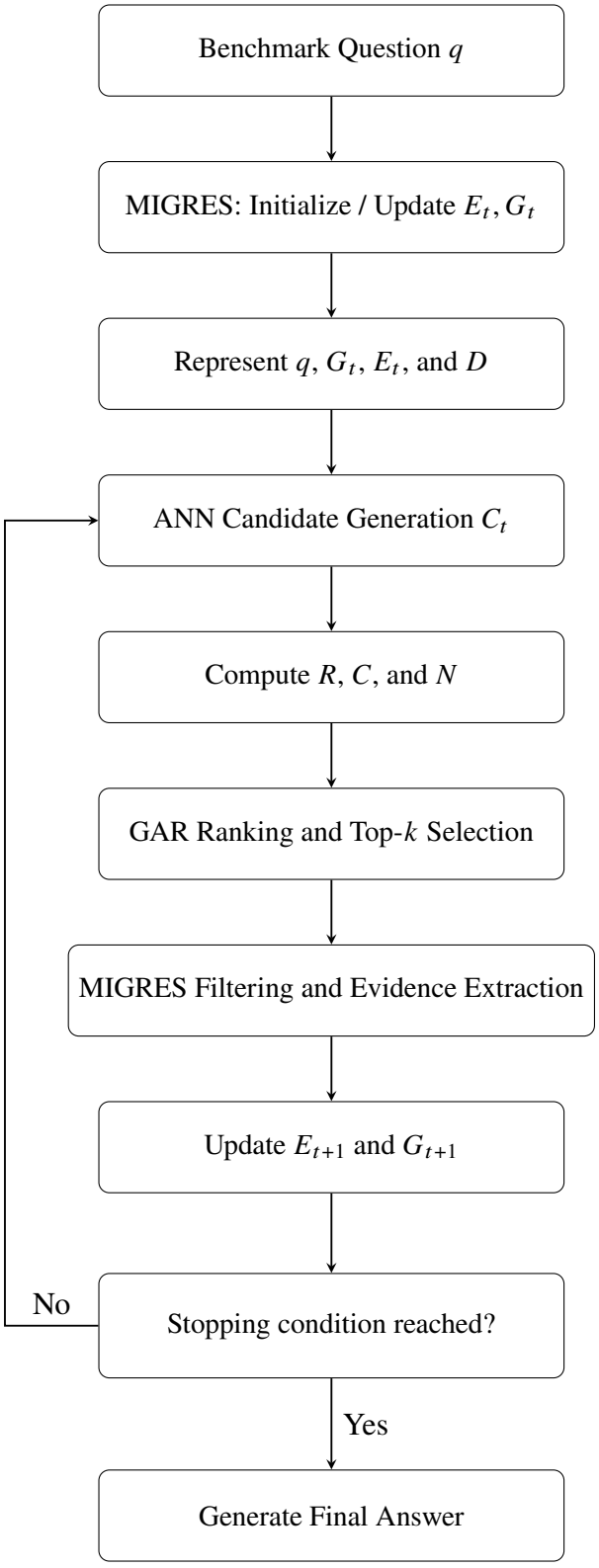


Figure 5.2: Operational flow of the proposed GAR-based iterative retrieval solution.

5.2 Mathematical Formulation

At each retrieval iteration t, GAR receives the question q, the previously collected evidence E_t, the current missing-information requirements $G_t = \{g_1, \ldots, g_m\}$, and the document corpus D:

$$(q, E_t, G_t, D).$$

Here, q represents the original question, E_t represents the evidence collected so far, G_t represents the information that is still missing, and D is the searchable document collection. The information gap and evidence are provided by the surrounding iterative retrieval process.

GAR uses a fixed sentence-embedding function $\phi(x)$ to represent questions, gaps, documents, and evidence as vectors. The primary configuration uses **sentence-transformers/multi-qa-mpnet-base-cos-v1** [6, 7]. The similarity between two text items is measured using cosine similarity:

$$s(x, y) = \frac{1}{2}\left(1 + \frac{\phi(x)^T \phi(y)}{\|\phi(x)\|_2 \|\phi(y)\|_2}\right).$$

This normalizes the similarity score to the range $[0, 1]$.

Before ranking, GAR creates a candidate set using approximate nearest-neighbor (ANN) search:

$$C_t = \mathrm{ANN}_{K_0}(q, D) \cup \bigcup_{i=1}^{m} \mathrm{ANN}_{K_0}(g_i, D),$$

where K_0 is the number of candidate documents retrieved for scoring. Duplicate documents are removed from C_t.

GAR then evaluates each candidate using three scores. First, gap relevance measures how relevant a document is to both the question and the current information gap. With equal gap weights, it is defined as:

$$R(d, q, G_t) = \lambda s(d, q) + (1 - \lambda)\frac{1}{m}\sum_{i=1}^{m} s(d, g_i),$$

where $0 \leq \lambda \leq 1$ controls the balance between the original question and the information gap.

Second, gap coverage measures how much of the missing information is covered by a document. Using normalized content tokens $\tau(\cdot)$:

$$C(d, G_t) = \frac{1}{m}\sum_{i=1}^{m} \frac{|\tau(d) \cap \tau(g_i)|}{|\tau(g_i)|}.$$

This score lies between 0 and 1. Empty gap requirements are removed before this calculation.

Third, evidence novelty measures how different a candidate document is from the evidence already collected:

$$N(d, E_t) = 1 - \max_{e \in E_t} s(d, e).$$

A document that is highly similar to existing evidence receives a lower novelty score. At the first iteration, where $E_0 = \emptyset$, novelty is set to 1.

The three scores are combined into the final GAR score:

$$S(d \mid q, E_t, G_t) = \alpha R(d, q, G_t) + \beta C(d, G_t) + \gamma N(d, E_t),$$

where

$$\alpha, \beta, \gamma \geq 0, \qquad \alpha + \beta + \gamma = 1.$$

The weights determine the importance of relevance, coverage, and novelty, respectively. Since all three component scores are normalized, the final GAR score also lies between 0 and 1.

Finally, GAR ranks the candidate documents according to their scores and returns the top-k documents:

$$T_t = \mathrm{TopK}_{d \in C_t} S(d \mid q, E_t, G_t).$$

The selected documents are then used in the next iteration, where the evidence and missing-information state may be updated before GAR performs the next retrieval.

5.3 Datasets and Model

5.3.1 Datasets

The evaluation uses six question-answering datasets that involve multi-hop reasoning, information gathering, or retrieval from supporting evidence. These datasets provide different types of information-seeking tasks and allow the proposed retrieval approach to be evaluated across varied retrieval and reasoning requirements.

The datasets used in this study are:

- **WikiHop:** A multi-document question-answering dataset that requires information to be combined across multiple documents [8].
- **HotpotQA:** A multi-hop question-answering dataset with supporting facts. The FullWiki setting is used to evaluate retrieval from a larger Wikipedia corpus [9].
- **MuSiQue:** A multi-hop question-answering dataset designed for compositional reasoning, with questions requiring two to four reasoning hops [10].
- **Natural Questions:** A question-answering dataset based on real search queries, with annotations derived from Wikipedia [11].
- **TriviaQA:** A question-answering dataset containing questions paired with independently collected evidence documents [12].
- **StrategyQA:** A question-answering dataset that requires implicit reasoning and uses supporting Wikipedia evidence [13].

Where official splits are available, the training, validation, and test sets are kept separate. The validation set is used for GAR hyperparameter selection, while the test set is reserved for final evaluation.

5.3.2 Model

GAR uses a Sentence Transformer model to convert questions, information gaps, documents, and collected evidence into numerical vector representations. Sentence Transformers are designed to generate embeddings that capture the semantic meaning of text, allowing text items to be compared based on semantic similarity rather than only exact word matching.

The primary model used in this study is `multi-qa-mpnet-base-cos-v1` from Sentence Transformers. It produces 768-dimensional dense embeddings and is designed for semantic search [6, 7]. The same model is used for questions, gap requirements, documents, and evidence to maintain a consistent representation throughout the retrieval process.

5.4 Baseline Comparison

The proposed method is compared with established retrieval approaches to evaluate its effectiveness. The main retrieval systems are summarized in Table 5.1.

Table 5.1: Retrieval baseline comparison

System	**Retrieval type**	**Role**
MIGRES+BM25	Sparse lexical	Main baseline [14]
MIGRES+DPR	Dense bi-encoder	Supervised dense baseline [15]
MIGRES+Contriever	Unsupervised dense	Contrastive-learning dense baseline [16]
MIGRES+ColBERTv2	Late interaction	Multi-vector retrieval baseline [5]
MIGRES+SPLADE	Neural sparse	Neural sparse baseline [17]
MIGRES+GAR	Gap-aware	Proposed retrieval method

For a fair comparison, the same dataset splits, questions, document corpus, retrieval depth k, and other system settings are used for all methods wherever possible. Only the retrieval method is changed. The baseline methods are used with their standard implementations and settings.

5.5 Experimental Design

The experiments are conducted in four main stages. First, the GAR hyperparameters are selected using the validation set. The selected configuration is then fixed and used for the main comparison with existing retrieval methods. Ablation experiments are then performed to study the contribution of the individual GAR components. Finally, the effect of the ANN candidate-pool size is analyzed.

5.5.1 Hyperparameter Selection

GAR uses five hyperparameters: α, β, γ, λ, and K_0. The roles of these parameters are shown in Table 5.2.

Table 5.2: GAR hyperparameters

Parameter	**Role**	**Description**
α	Relevance weight	Controls the importance of relevance.
β	Coverage weight	Controls the importance of coverage.
γ	Novelty weight	Controls the importance of new information.
λ	Question weight	Controls the importance of the original question.
K_0	Candidate-pool size	Controls the number of candidate documents.

The validation set is used to select the GAR configuration:

$$\theta_{\mathrm{GAR}} = \{\alpha, \beta, \gamma, \lambda, K_0\}.$$

Bayesian optimization is used to search for the configuration that gives the best validation performance [18]. Recall@k is used as the primary validation metric. The search includes different values of α, β, γ, and λ, while K_0 is selected from the predefined values:

$$K_0 \in \{50, 100, 200, 500\}.$$

The best configuration is selected from the validation results and then fixed for the remaining experiments. The test set is not used to change the selected hyperparameters.

5.5.2 Main Comparison

After the GAR configuration is fixed, the main experiment compares GAR with established retrieval methods. The methods evaluated are BM25, DPR, Contriever, ColBERTv2, and SPLADE.

All methods use the same datasets, questions, document corpus, and final retrieval depth k wherever possible. This allows the comparison to focus on the difference between the retrieval methods.

The main evaluation measures retrieval performance and downstream answer quality.

5.5.3 Ablation Study

An ablation study is conducted to determine the contribution of the three main GAR components: relevance, coverage, and novelty.

The complete GAR score is:

$$S_{\mathrm{GAR}} = \alpha R + \beta C + \gamma N.$$

Each component is removed separately to create three variants:

$$S_{-R} = \beta C + \gamma N,$$

$$S_{-C} = \alpha R + \gamma N,$$

$$S_{-N} = \alpha R + \beta C.$$

The results of these variants are compared with the complete GAR model to determine the contribution of each component.

5.5.4 Candidate-Pool Analysis

The effect of the ANN candidate-pool size K_0 is also evaluated. The following values are considered:

$$K_0 \in \{50, 100, 200, 500\}.$$

The different settings are compared based on retrieval performance and latency. The validation results are used to select the final K_0, which is then fixed for the test evaluation.

5.5.5 Evaluation Metrics

Retrieval Evaluation

The retrieval performance is evaluated using Recall@k, Precision@k, Mean Reciprocal Rank (MRR), and nDCG@k. Recall@k measures how much of the relevant evidence is retrieved in the first k documents. Precision@k measures how many of the retrieved documents are relevant. MRR measures how early the first relevant document appears, while nDCG@k measures the quality of the document ranking when graded relevance is available [19]. Evidence coverage is also reported where the dataset provides suitable annotations.

Downstream Answer Evaluation

The final answers are evaluated using the official metrics provided by each dataset. These include Exact Match (EM), token-level F1, classification accuracy, and supporting-evidence quality where applicable. Official evaluation scripts are used whenever available.

Efficiency Evaluation

The efficiency of the retrieval methods is evaluated using retrieval latency, ANN indexing time, index memory, number of retrieval iterations, and number of documents scored by GAR. The complete system also records the number of LLM calls, input and output tokens, and API cost where applicable.

Statistical Analysis

The results are reported with effect sizes and confidence intervals. Statistical significance is tested using a paired statistical procedure, such as paired bootstrap or paired permutation testing, for per-question metrics. The statistical procedure is selected before the final test results are examined.

5.6 Experimental Environment and Reproducibility

The experiments use a software environment that supports dataset preparation, sentence embedding, ANN indexing, GAR retrieval, and evaluation. The exact software versions, hardware configuration, and model settings used in the experiments are recorded.

For reproducibility, the datasets, document corpus, models, GAR parameters, ANN settings, and software versions used in the experiments are recorded. The same data and evaluation settings are used for all retrieval methods. The GAR parameters selected during validation are fixed before the final test.

The experimental pipeline is:

$$\text{Dataset} \rightarrow \text{MIGRES State} \rightarrow \text{GAR Retrieval and Ranking} \rightarrow \text{Evidence Update} \rightarrow \text{Evaluation}.$$

5.7 Chapter Solution Summary

This chapter presented the proposed Gap-Aware Retrieval (GAR) approach for iterative document retrieval. GAR considers the original question, the current information gap, and previously collected evidence to select relevant and less redundant documents.

The chapter described the GAR architecture, retrieval process, mathematical formulation, and experimental design. GAR will evaluated against established retrieval methods using controlled experiments, with hyperparameters selected on the validation set before final test evaluation. Ablation and efficiency analyses are also included to examine the contribution and practical performance of the proposed approach.

Overall, the methodology provides a complete framework for implementing and evaluating GAR for iterative information retrieval.

CHAPTER 6

SOFTWARE AND HARDWARE REQUIREMENTS

The proposed solution requires a Python-based environment for dataset preparation, embedding generation, vector search, GAR retrieval and ranking, hyperparameter selection, and evaluation. Jupyter Notebook is used for experiment development and analysis, while LaTeX is used for preparing the research document. The main software and hardware requirements are described below.

6.1 Software Requirements

The main software tools required for the implementation are shown in Table 6.1.

Table 6.1: Software requirements

Software / Tool	**Version**	**Purpose**
Python	3.14.7	Main programming environment for GAR implementation, data processing, retrieval, and evaluation.
Jupyter Notebook	7.6.2	Experiment development, execution, and result analysis.
PyTorch	2.13.0	Tensor computation and neural model execution.
Sentence Transformers	5.6.1 (model: multi-qa-mpnet-base-cos-v1)	Generates semantic embeddings for questions, documents, gaps, and evidence.
Pinecone	Client v9.1.0 (API 2025-10)	Stores document embeddings and performs vector similarity search for candidate retrieval.
Bayesian Optimization Library	3.3.0 (bayesian-optimization)	Optimizes GAR hyperparameters on the validation set.
LaTeX	Overleaf	Preparation of the research document.

The primary embedding model is `multi-qa-mpnet-base-cos-v1` from Sentence Transformers. It is used to generate a common semantic representation for questions, information gaps, documents, and collected evidence. The model remains fixed and is not fine-tuned during the primary experiments.

6.2 Hardware Requirements

The main hardware requirements are shown in Table 6.2.

Table 6.2: Hardware requirements

Hardware	**Purpose**
CPU	Required for data preprocessing, experiment execution, GAR scoring, and evaluation.
RAM	Required for loading datasets, processing documents, and handling embedding and experiment data.
GPU	Recommended for faster embedding generation and execution of neural retrieval models.
Storage	Required for datasets, document files, model files, and experiment results.
Internet Connection	Required for downloading datasets and pretrained models and for accessing Pinecone.

CHAPTER 7

APPLICATIONS

1. **Multi-Hop Question Answering:** The methodology is suited to questions whose answers require complementary information from multiple documents or retrieval steps.

2. **Iterative Retrieval-Augmented Generation:** GAR can serve as a retrieval component in RAG systems that repeatedly acquire evidence as the answer is developed and the information requirement changes.

3. **Research-Oriented Evidence Retrieval:** The gap and evidence signals can help research-oriented systems focus subsequent retrieval on information that is still required rather than repeatedly collecting already-covered material.

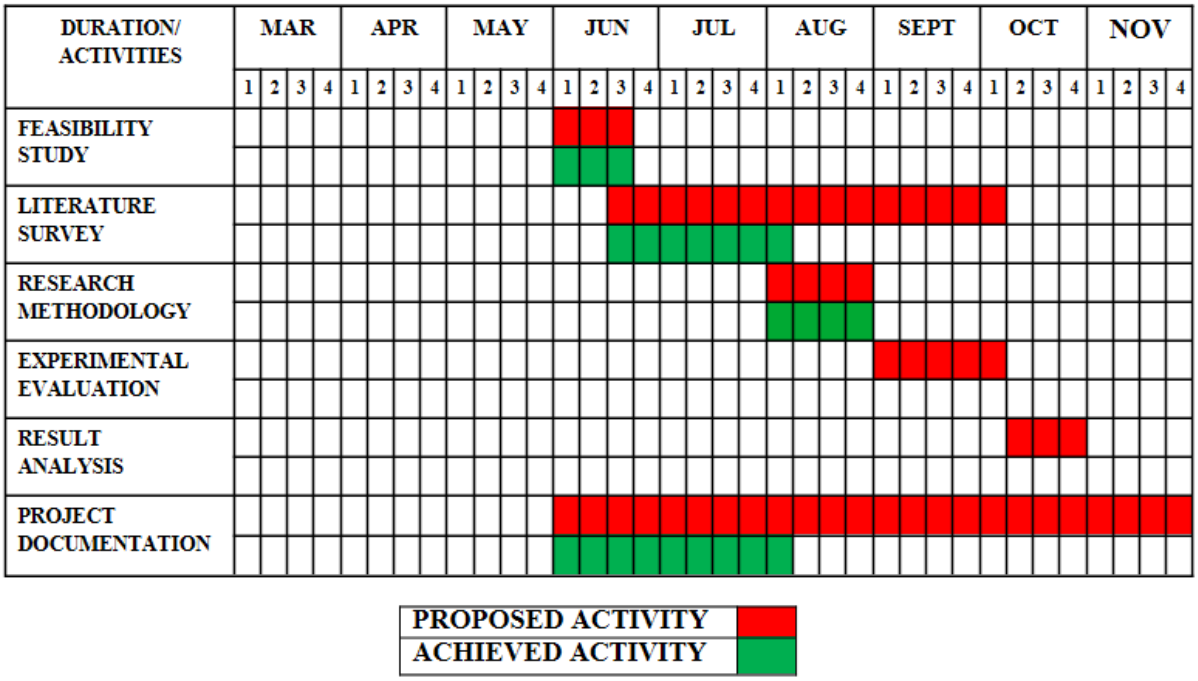
4. **Large-Scale Evidence Gathering:** ANN candidate generation allows the retrieval stage to operate over large document collections while restricting full GAR scoring to a manageable candidate set.

5. **Missing-Information-Guided AI Systems:** The approach is applicable to systems that explicitly maintain an information-gap state and can provide that state to the retrieval component.

CHAPTER 8

GANTT CHART

Figure 6.1: Gantt Chart



CHAPTER 9

CONCLUSION

This methodology presents a controlled approach for evaluating Gap-Aware Retrieval (GAR) within the MIGRES framework. GAR uses the current information gap and previously collected evidence to retrieve and rank documents based on relevance, coverage, and novelty. The selected documents are then passed to MIGRES for filtering, extraction, and further iteration.

GAR is evaluated against established retrieval methods, including BM25, DPR, Contriever, ColBERTv2, and SPLADE. Ablation studies are used to examine the contribution of each GAR component, while retrieval quality, answer quality, and efficiency are evaluated using predefined metrics.

Overall, the methodology is designed to determine whether using the information gap and existing evidence can improve retrieval in iterative information-seeking systems while keeping the surrounding framework unchanged.

REFERENCES

[1] K. Wang, F. Duan, P. Li, S. Wang, and X. Cai, "Llms know what they need: Leveraging a missing information guided framework to empower retrieval-augmented generation," in *Proceedings of the 31st International Conference on Computational Linguistics*, 2025, pp. 2379–2400.

[2] M. Pickett, J. Hartman, A. K. Bhowmick, R. ul Alam, and A. Vempaty, "Better rag using relevant information gain," *arXiv preprint arXiv:2407.12101*, 2024.

[3] Z. Wang, Z. Liang, Z. Shao, Y. Ma, H. Dai, B. Chen, L. Mao, C. Lei, Y. Ding, and H. Li, "Infogain-rag: Boosting retrieval-augmented generation through document information gain-based reranking and filtering," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 7190–7204.

[4] J. Hwang, J. Park, H. Park, D. Kim, S. Park, and J. Ok, "Retrieval-augmented generation with estimation of source reliability," in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025, pp. 34 279–34 303.

[5] K. Santhanam, O. Khattab, J. Saad-Falcon, C. Potts, and M. Zaharia, "Colbertv2: Effective and efficient retrieval via lightweight late interaction," in *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*, 2022, pp. 3715–3734.

[6] N. Reimers and I. Gurevych, "Sentence-bert: Sentence embeddings using siamese bert-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing*, 2019, pp. 3982–3992.

[7] Sentence Transformers, "multi-qa-mpnet-base-cos-v1," Hugging Face model card, 2020, 768-dimensional sentence-transformers model designed for semantic search.

[8] J. Welbl, P. Stenetorp, and S. Riedel, "Constructing datasets for multi-hop reading comprehension across documents," *Transactions of the Association for Computational Linguistics*, vol. 6, pp. 287–302, 2018.

[9] Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. Cohen, R. Salakhutdinov, and C. D. Manning, "Hotpotqa: A dataset for diverse, explainable multi-hop question answering," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, 2018, pp. 2369–2380.

[10] H. Trivedi, N. Balasubramanian, T. Khot, and A. Sabharwal, "Musique: Multihop questions via single-hop question composition," *Transactions of the Association for Computational Linguistics*, vol. 10, pp. 539–554, 2022.

[11] T. Kwiatkowski, J. Palomaki, O. Redfield, M. Collins, A. Parikh, C. Alberti, D. Epstein, I. Polosukhin, J. Devlin, K. Lee, K. Toutanova, L. Jones, M. Kelcey, M.-W. Chang, J. Uszkoreit, Q. Le, and S. Petrov, "Natural questions: A benchmark for question answering research," *Transactions of the Association for Computational Linguistics*, vol. 7, pp. 452–466, 2019.

[12] M. Joshi, E. Choi, D. Weld, and L. Zettlemoyer, "Triviaqa: A large scale distantly supervised challenge dataset for reading comprehension," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics*, 2017, pp. 1601–1611.

[13] M. Geva, D. Khashabi, E. Segal, T. Khot, D. Roth, and J. Berant, "Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies," *Transactions of the Association for Computational Linguistics*, vol. 9, pp. 346–361, 2021.

[14] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: Bm25 and beyond," *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.

[15] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, 2020, pp. 6769–6781.

[16] G. Izacard, M. Caron, L. Hosseini, S. Riedel, P. Bojanowski, A. Joulin, and E. Grave, "Unsupervised dense information retrieval with contrastive learning," in *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, 2022, pp. 2665–2679.

[17] T. Formal, C. Lassance, B. Piwowarski, and S. Clinchant, "Splade: Sparse lexical and expansion model for first stage ranking," in *arXiv preprint arXiv:2107.05720*, 2021.

[18] J. Snoek, H. Larochelle, and R. P. Adams, "Practical bayesian optimization of machine learning algorithms," in *Advances in Neural Information Processing Systems*, vol. 25, 2012.

[19] K. Järvelin and J. Kekäläinen, "Cumulated gain-based evaluation of ir techniques," *ACM Transactions on Information Systems*, vol. 20, no. 4, pp. 422–446, 2002.